Lecture 20:
Simulation-Based Inference

November 2, 2026

simulation-based inference using ABC — simulator $F(\theta)$ instead of the explicit likelihood

$$\mathscr{D}' \sim F(\theta)$$

simulator F simulates the data generation process

see ABC unit

simulation-based inference using ABC — simulator $F(\theta)$ instead of the explicit likelihood

$$\mathscr{D}' \sim F(\theta)$$

simulator F simulates the **data generation process**

including noise, systematics, etc

see ABC unit

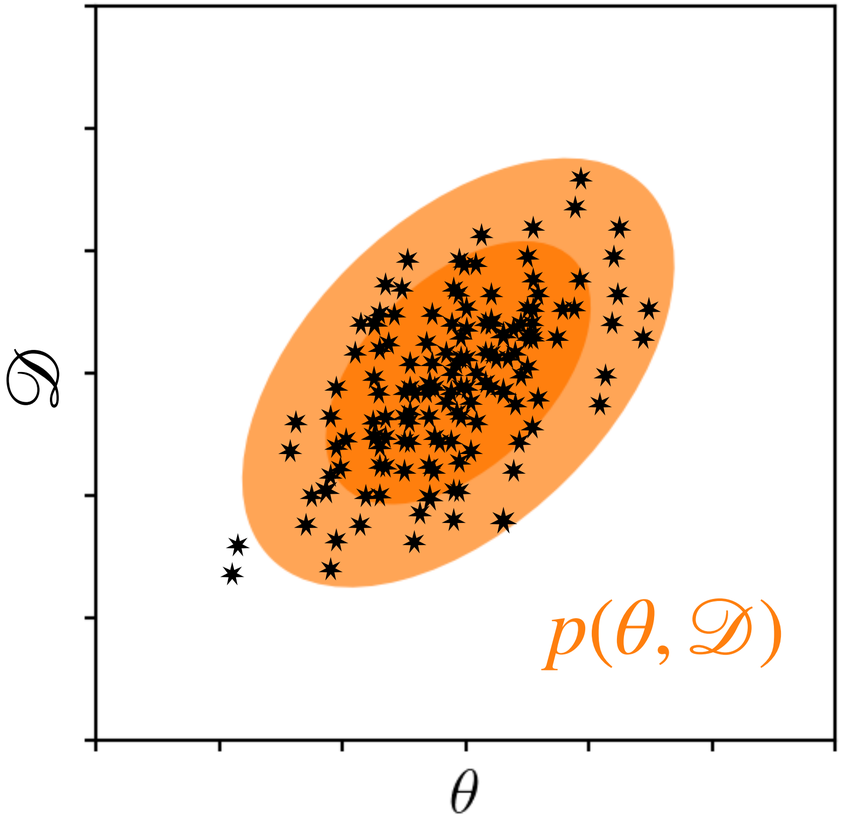
simulation-based inference using ABC — simulator $F(\theta)$ instead of the explicit likelihood

$$\mathscr{D}' \sim F(\theta) \quad \approx \quad \mathscr{D}' \sim p(\mathscr{D} \mid \theta)$$

simulator F approximates sampling the likelihood

with a prior $p(\theta)$ and F, we can sample the joint distribution $p(\theta, \mathscr{D})$

$$\theta' \sim p(\theta) \quad \mathscr{D}' \sim F(\theta')$$



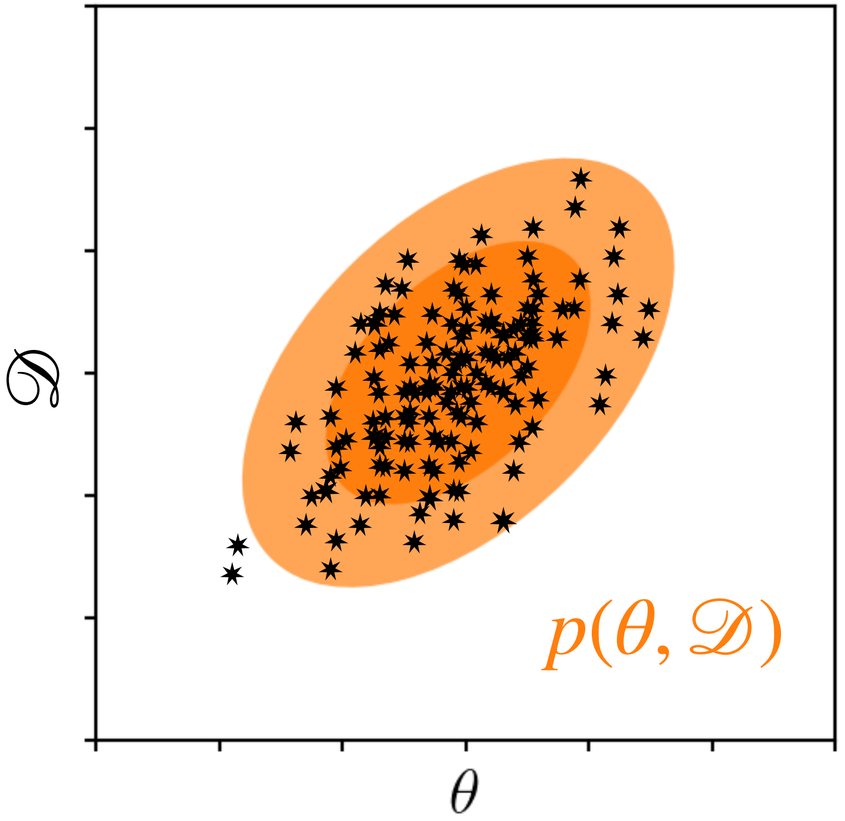
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approximate bayesian computation

$$p(\theta \mid \mathscr{D}_{\text{obs}}) \approx p(\theta \mid \mathscr{D} \approx \mathscr{D}_{\text{obs}})$$

$$\approx p(\theta \mid \rho(\mathscr{D}_{\text{obs}}, \mathscr{D}) < \epsilon)$$



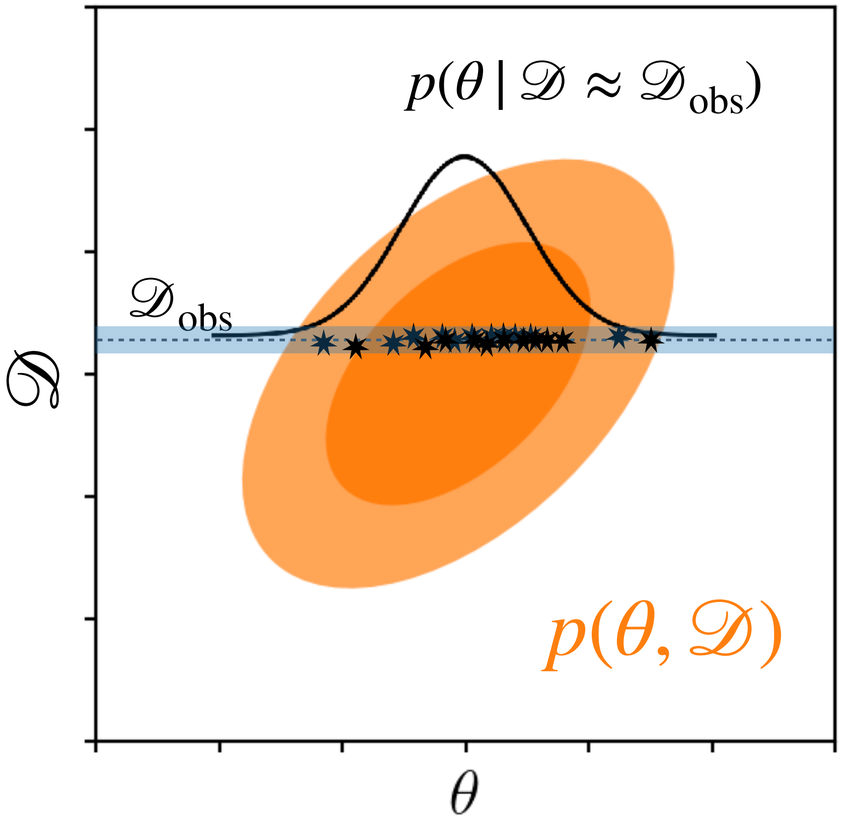
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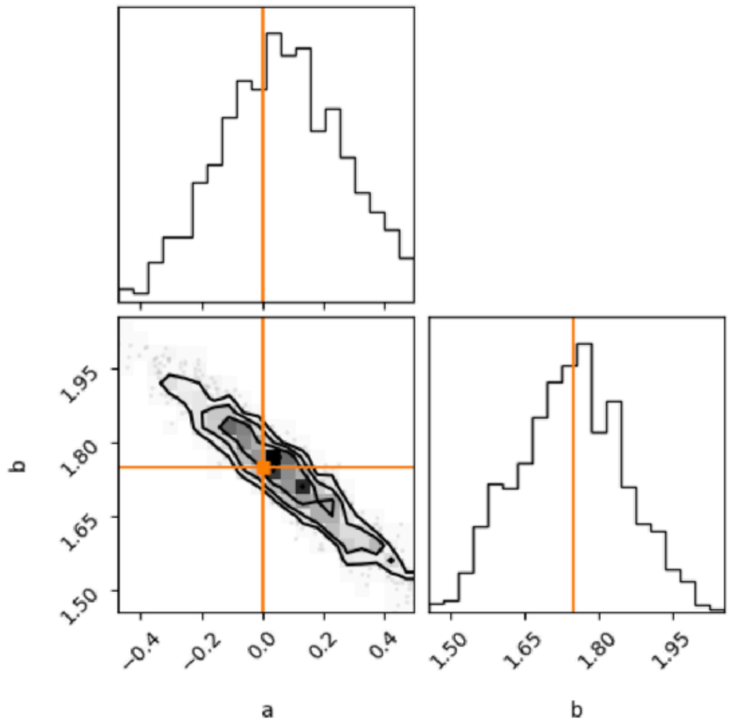


ABC is **inefficient**

In [85]:
```
epsilon = 0.2
print('reject %f percent of simulations' % (100.*(1-np.mean(rho < epsilon))))
```

```
reject 99.809800 percent of simulations
```

In [83]:
```
fig = corner.corner(samples[:,:2][rho < epsilon], labels=['a', 'b'])
corner.overplot_lines(fig, [a_true, b_true], color="C1")
corner.overplot_points(fig, np.array([a_true, b_true])[None], marker="s", color="C1")
```



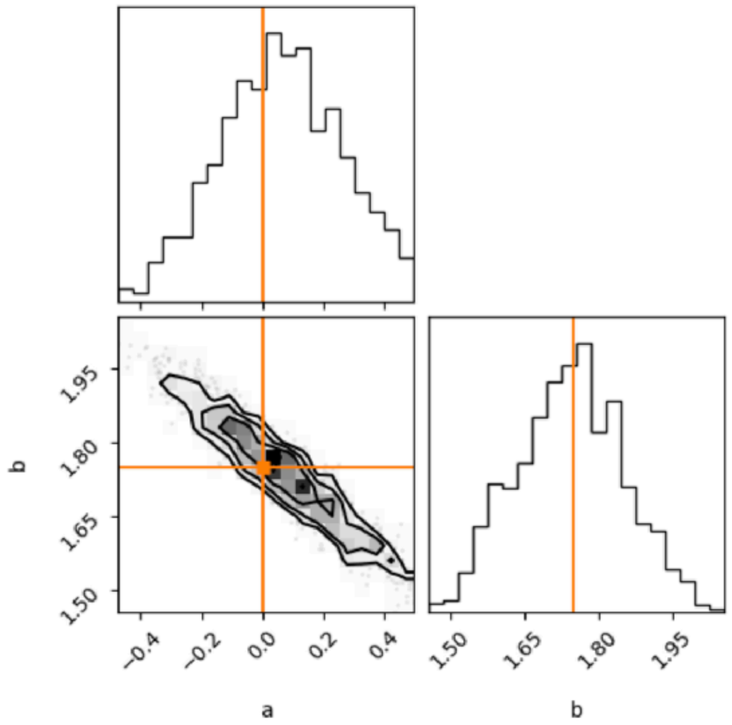
see ABC unit jupyter notebook

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see ABC unit jupyter notebook

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reject 99.809800 percent of simulations

In [83]: fig = corne
         corner.over
         corner.over
```


GalSBI: phenomenological galaxy population model for cosmology using simulation-based inference

Silvan Fischbacher[a,*] Tomasz Kacprzak[b] Luca Tortorelli[c] Beatrice Moser[a] Alexandre Refregier[a] Patrick Gebhardt[c] and Daniel Gruen[c,d]

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ABSTRACT: We present GalSBI, a phenomenological model of the galaxy population for cosmological applications using simulation-based inference. The model is based on analytical parametrizations of galaxy luminosity functions, morphologies and spectral energy distributions. Model constraints are derived through iterative Approximate Bayesian Computation, by comparing Hyper Suprime-Cam deep field images with simulations which include a forward model of instrumental, observational and source extraction effects. We developed an emulator trained on image simulations using a normalizing flow. We use it to accelerate the inference by predicting detection probabilities, including blending effects and photometric properties of each object, while accounting for background and PSF variations. This enables robustness tests for all elements of the forward model and the inference. The model demonstrates excellent performance when comparing photometric properties from simulations with observed imaging data for key parameters such as magnitudes, colors and sizes. The redshift distribution of simulated galaxies agrees well with high-precision photometric redshifts in the COSMOS field within 1.5σ for all magnitude cuts. Additionally, we demonstrate how GalSBI's redshifts can be utilized for splitting galaxy catalogs into tomographic bins, highlighting its potential

*Corresponding author.

JCAP06(2025)007

still used in the literature

ABC becomes **prohibitive** if simulator is expensive

```
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reject 99.809800 percent of simulations

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         corner.over
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Abstract: We present GalSBI, a phenomenological model of the galaxy population for cosmological applications using simulation-based inference. The model is based on analytical parametrizations of galaxy luminosity functions, morphologies and spectral energy distributions. Model constraints are derived through iterative Approximate Bayesian Computation, by comparing Hyper Suprime-Cam deep field images with simulations which include a forward model of instrumental, observational and source extraction effects. We developed an emulator trained on image simulations using a normalizing flow. We use it to accelerate the inference by predicting detection probabilities, including blending effects and photometric properties of each object, while accounting for background and PSF variations. This enables robustness tests for all elements of the forward model and the inference. The model demonstrates excellent performance when comparing photometric properties from simulations with observed imaging data for key parameters such as magnitudes, colors and sizes. The redshift distribution of simulated galaxies agrees well with high-precision photometric redshifts in the COSMOS field within 1.5σ for all magnitude cuts. Additionally, we demonstrate how GalSBI's redshifts can be utilized for splitting galaxy catalogs into tomographic bins, highlighting its potential

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still used in the literature

can we conduct more efficient simulation-based inference?

variational inference: optimize ϕ in a surrogate $q_\phi(x)$ to estimate a target $p^*(x)$

$$p^*(x) \approx q_\phi(x)$$

$$\min_\phi D_{\mathrm{KL}}\left(q_\phi(x) \parallel p^*(x)\right)$$

see variational inference unit

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$$p^*(x) \approx q_\phi(x)$$

$$\min_\phi D_{\mathrm{KL}}\left(q_\phi(x) \parallel p^*(x)\right)$$

Kullback-Leibler Divergence

$$D_{\mathrm{KL}} = \int \mathrm{d}x \; q_\phi(x) \; \log \frac{q_\phi(x)}{p(x)}$$

information theory: *(cross entropy) - (entropy)* or a "metric" discriminating information between two distributions

simulation-based inference (SBI) with VI

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optimize ϕ in a surrogate distribution $q_\phi(\theta, \mathscr{D})$ to estimate a target distribution $p(\theta, \mathscr{D})$

simulation-based inference (SBI) with VI

$$\min_{\phi} D_{\mathrm{KL}}\left(p(\theta, \mathscr{D}) \parallel q_{\phi}(\theta, \mathscr{D})\right) = \int \mathrm{d}\theta \; p(\theta, \mathscr{D}) \; \log \frac{p(\theta, \mathscr{D})}{q_{\phi}(\theta, \mathscr{D})}$$

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p and q_ϕ order is intentional

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p and q_ϕ order is intentional

without explicit likelihood, we **cannot** evaluate $p(\theta \mid \mathscr{D})$ or $p(\theta, \mathscr{D})$

but we can $(\theta', \mathscr{D}') \sim p(\mathscr{D} \mid \theta)$ using F

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$$= \int \mathrm{d}\theta \; p(\theta, \mathscr{D}) \; \left[\log p(\theta, \mathscr{D}) - \log q_{\phi}(\theta, \mathscr{D})\right]$$

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SBI: with $p(\theta)$ and F

$(\theta', \mathscr{D}') \sim p(\theta, \mathscr{D})$

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SBI: with $p(\theta)$ and F

$$(\theta', \mathscr{D}') \sim p(\theta, \mathscr{D})$$

Monte Carlo integration

$$\mathbb{E}_p(f) = \int f(x) \; p(x) \; \mathrm{d}x \approx \frac{1}{S} \sum_{x_i \sim p(x)}^{x_S} f(x_i)$$

simulation-based inference (SBI) with VI

$$\min_{\phi} D_{\mathrm{KL}}\left(p(\theta, \mathscr{D}) \parallel q_{\phi}(\theta, \mathscr{D})\right) = \int \mathrm{d}\theta \; p(\theta, \mathscr{D}) \; \log \frac{p(\theta, \mathscr{D})}{q_{\phi}(\theta, \mathscr{D})}$$

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doesn't depend on ϕ!

simulation-based inference (SBI) with VI

$$
\begin{aligned}
\min_{\phi} D_{\mathrm{KL}}\left(p(\theta, \mathscr{D}) \parallel q_{\phi}(\theta, \mathscr{D})\right) &= \int \mathrm{d}\theta \; p(\theta, \mathscr{D}) \; \log \frac{p(\theta, \mathscr{D})}{q_{\phi}(\theta, \mathscr{D})} \\
&= \int \mathrm{d}\theta \; p(\theta, \mathscr{D}) \; \left[\log p(\theta, \mathscr{D}) - \log q_{\phi}(\theta, \mathscr{D})\right] \\
&\approx \sum_{(\theta', \mathscr{D}') \sim p(\theta, \mathscr{D})} \left[\log p(\theta', \mathscr{D}') - \log q_{\phi}(\theta', \mathscr{D}')\right] \\
&\equiv \sum_{(\theta', \mathscr{D}') \sim p(\theta, \mathscr{D})} \left[-\log q_{\phi}(\theta', \mathscr{D}')\right]
\end{aligned}
$$

simulation-based inference (SBI) with VI

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does not require evaluating $p(\theta, \mathscr{D})$ — only $q_{\phi}(\theta, \mathscr{D})$

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does not require evaluating $p(\theta, \mathscr{D})$ — only $q_{\phi}(\theta, \mathscr{D})$

if $q_{\phi}(\theta, \mathscr{D})$ is easy to evaluate, optimization is easy

SBI + VI — Bayesian workflow

1. sample $\{(\theta', \mathscr{D}')\} \sim p(\theta, \mathscr{D})$ using simulator F

SBI + VI — Bayesian workflow

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2. select $q_\phi(\theta, \mathscr{D})$

SBI + VI — Bayesian workflow

1. sample $\{(\theta', \mathscr{D}')\} \sim p(\theta, \mathscr{D})$ using simulator F

2. select $q_\phi(\theta, \mathscr{D})$ and optimize ϕ

$$\min_\phi D_{\mathrm{KL}}\left(p(\theta, \mathscr{D}) \parallel q_\phi(\theta, \mathscr{D})\right) \approx \sum_{(\theta', \mathscr{D}') \sim p(\theta, \mathscr{D})} \left[-\log q_\phi(\theta', \mathscr{D}')\right]$$

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$$p(\theta, \mathscr{D}) \approx q_\phi(\theta, \mathscr{D})$$

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3. ABC using surrogate q_ϕ

SBI + VI — Bayesian workflow

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3. ABC using surrogate q_ϕ

$$(\theta', \mathscr{D}') \sim q_\phi(\theta, \mathscr{D})$$

we can sample the joint distribution **without** running an expensive simulator

groups:

Group 1: Ines, Jayden, Asa

Group 2: Madeline, Alexander, Noah

Group 3: Nikhil, Jules, Michelle, Kaleo

Group 4: Tae Bong, Nicholas, Olivia

Group 5: Bella, Bocheng, Shreya, John

in-class: SBI with variational inference and in-class exercise

ast384c/inclass/lec20_sbi_intro/sbi_vi.ipynb